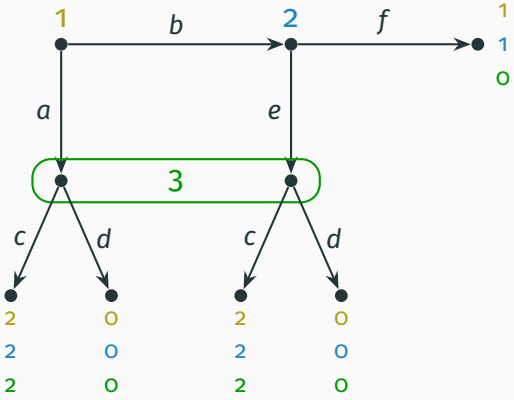
Chapter 11: Weak Sequential Equilibrium

ECON 5001
Prof. Paul J. Healy
Based on: *Game Theory* by Giacomo Bonanno

Updated 2026-11-09 at 21:12:30

Where We Left Off: Selten's Horse



Four pure NE. All are SPNE:

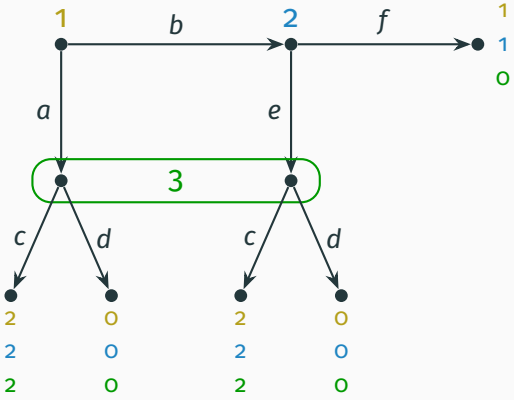
(a, e, c) (a, f, c)

(b, e, c) (b, f, d)

SPNE only checks **subgames**

- P3's info set isn't a subgame
- So bad choices there go unchecked

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Fix: give P3 a **belief** about where she is. Check every choice against it

MobLab: “Herding (Information Cascade)”

- MobLab time!
- A jar is either mostly **Blue** or mostly **Red** (equally likely)
  - Mostly $=$ 60% of the balls
- You privately draw one ball (it goes back in)
- One at a time, in random order, guess: Blue jar or Red jar?
  - You see everyone’s **guesses** before you. Not their balls
- Paid for a correct guess

Feel free to talk! And play it the way you’d actually play it.

Beliefs at Information Sets

Example: The Donut Truck

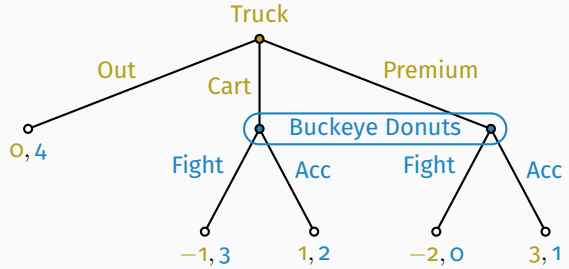
A new donut truck thinks about parking near Buckeye Donuts on High St

- **Out**: stay away
- **Cart**: a cheap little cart (weak)
- **Premium**: a full truck with a big menu (strong)

Buckeye Donuts sees **that** the truck showed up, not **which** plan

- **Fight**: price war. **Acc**: accommodate
- Fighting a weak cart pays. Fighting a strong truck doesn't

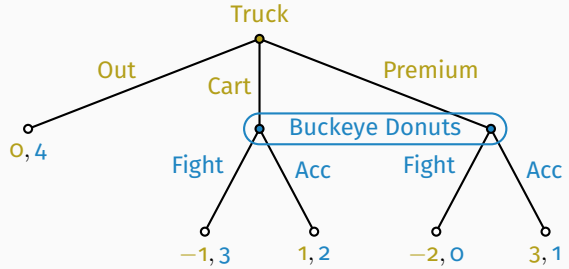
The Donut Truck



Truck \ Buckeye	Fight	Acc
Out	0, 4	0, 4
Cart	−1, 3	1, 2
Premium	−2, 0	3, 1

Proper subgames? _____ So $SPNE = NE$

The Donut Truck

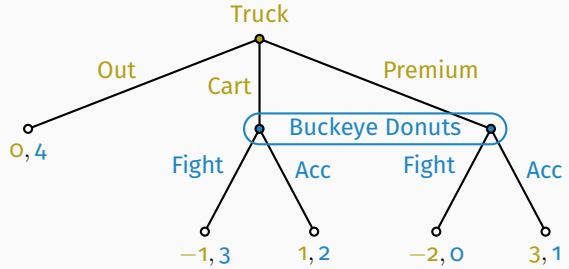


Truck \ Buckeye	Fight	Acc
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NE: _____ _____

The Donut Truck



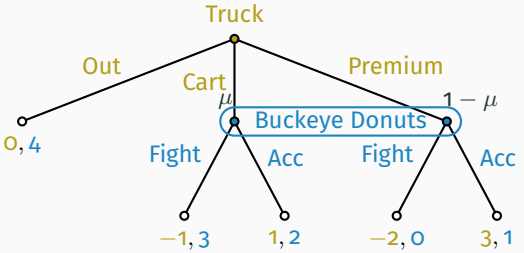
Truck \ Buckeye	Fight	Acc
Out	0, 4	0, 4
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Proper subgames? _____ So *SPNE* = *NE*

NE: _____ _____

Is (Out, Fight) reasonable? **It depends on what Buckeye believes**

What Should Buckeye Believe?



$\mu = \Pr(\text{Cart} \mid \text{truck showed up})$

Fight: _____

Acc: _____

In words: fight when the truck is likely enough to be a weak cart

$$\underbrace{3\mu}_{\text{payoff from Fight}} \geq \underbrace{2\mu + 1(1-\mu)}_{\text{payoff from Acc}} \iff \mu \geq _____$$

Assessment

Definition (Bonanno 11.1.1)
An **assessment** (σ, μ) is

- σ: a (behavioral) strategy profile, and
- μ: a **system of beliefs**. One distribution over the nodes in each info set

μ: "if I ever got here, where would I think I am?"

- Even at information sets that σ never reaches!
- Singleton info sets: belief is 1 on that node. We'll skip writing those

Ex: $\sigma =$ (Out, Fight), $\mu =$ (Cart: $\frac{2}{3}$, Premium: $\frac{1}{3}$)

Two Requirements

1. Sequential rationality (Bonanno 11.1.2)

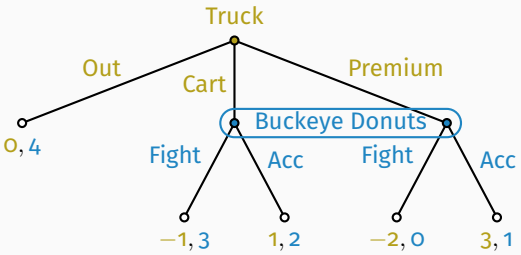
- At **every** info set, reached or not:
- Your choice maximizes expected payoff, given your beliefs μ there and everyone's strategies from then on

2. Bayesian updating (Bonanno 11.2.2)

- If σ reaches an info set with positive probability: beliefs come from σ by Bayes' rule
- If σ never reaches it: **any** beliefs are OK

Sequential rationality also means you can't gain by changing your choice here **and** later choices too.

Bayesian Updating: Example



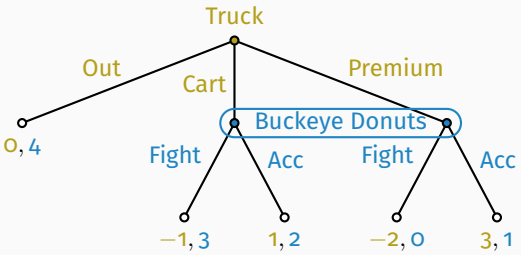
Truck mixes:

$$\begin{pmatrix} \text{Out} & \text{Cart} & \text{Premium} \\ \frac{1}{2} & \frac{1}{10} & \frac{2}{5} \end{pmatrix}$$

Pr(Buckeye's info set reached) $=$ _____

$\mu = P(\text{Cart} \mid \text{reached}) =$ _____

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Pr(Buckeye's info set reached) = _____

$\mu = P(\text{Cart} \mid \text{reached}) =$ _____

If the truck plays **Out** for sure: info set never reached
$\Rightarrow$ Bayes' rule says nothing ($\frac{0}{0}$). Any μ allowed

Weak Sequential Equilibrium

Definition (Bonanno 11.3.1)
A **weak sequential equilibrium** (WSE) is an assessment (σ, μ) that satisfies

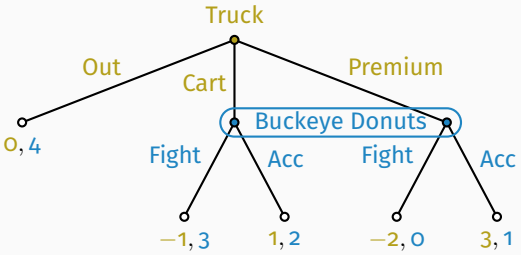
1. Sequential rationality, and
2. Bayesian updating at reachable information sets

How to find them:

1. Delete choices that are strictly dominated at an info set (bad for **any** belief)
2. Guess the first mover's strategy
3. Bayes' rule gives beliefs where it can
4. Check sequential rationality everywhere. Repeat

Also called "weak perfect Bayesian equilibrium" in other books.

Donut Truck: WSE #1



Try $\sigma =$ (Premium, Acc)

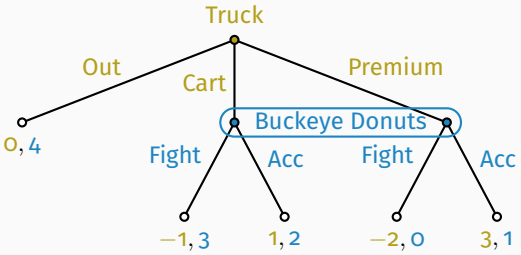
Info set reached? _____

Bayes: $\mu =$ _____

Buckeye: Acc optimal given μ? _____

Truck: Premium optimal? _____

Donut Truck: WSE #1



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Info set reached? _____

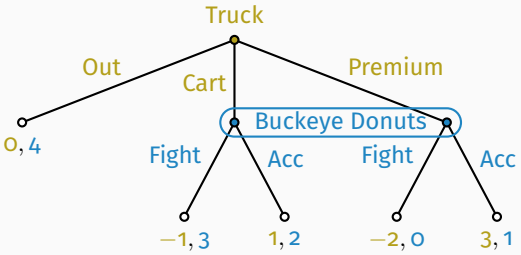
Bayes: $\mu =$ _____

Buckeye: Acc optimal given μ? _____

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WSE: (Premium, Acc) with $\mu =$ _____

Donut Truck: WSE #2?



Try $\sigma =$ (Out, Fight)

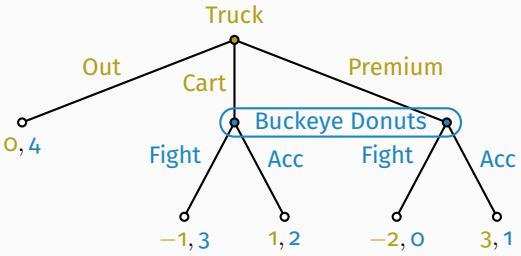
Info set reached? _____

So Bayes says: _____

Buckeye: Fight optimal iff _____

Truck: Out optimal? _____

Donut Truck: WSE #2?



Try $\sigma =$ (Out, Fight)

Info set reached? _____

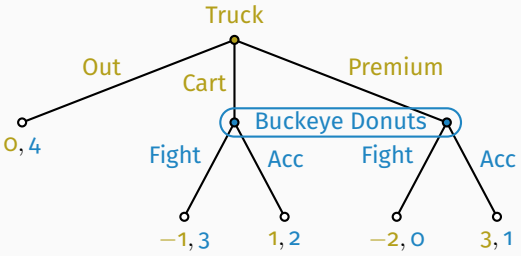
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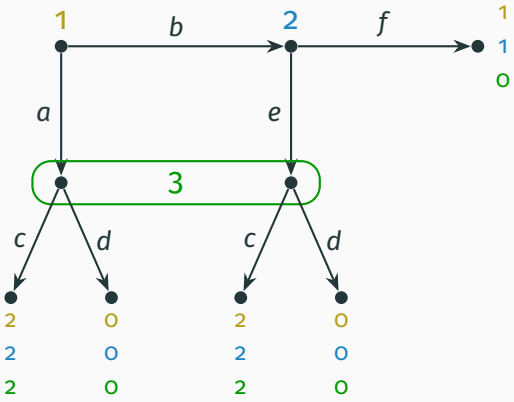
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The threat works only if Buckeye **believes** a surprise entrant is weak

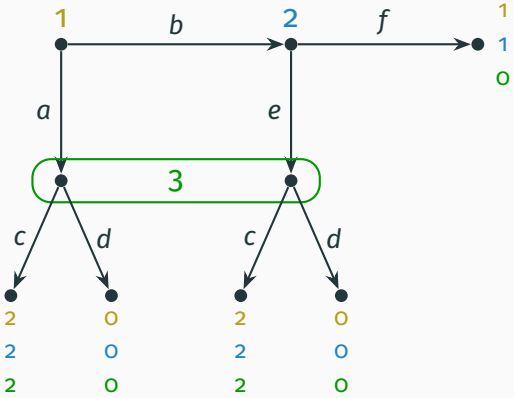
Back to Selten's Horse



Which NE can be part of a WSE?

NE	WSE?	Why / beliefs
(a, e, c)		
(a, f, c)		
(b, e, c)		
(b, f, d)		

Back to Selten's Horse

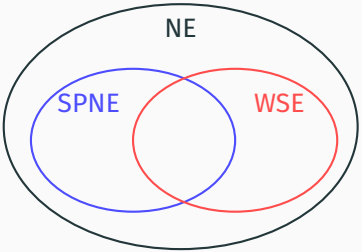


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Lesson: WSE kills the silly SPNE from Chapter 7

How WSE Relates to NE and SPNE



Theorem (11.3.1): WSE $\Rightarrow$ NE

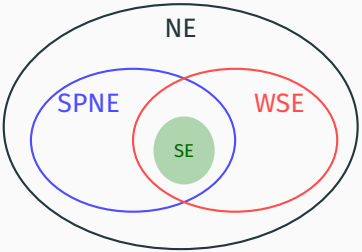
Theorem (11.3.2): every finite game has a WSE

But: a WSE need not be a SPNE!

- Silly beliefs at unreached info sets can support silly play inside a subgame

Bonanno Figures 11.10–11.11 give an example of a WSE that isn't a SPNE.

How WSE Relates to NE and SPNE



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Coming Attraction: Sequential Equilibrium

WSE's weak spot: **anything goes** off the path of play

Sequential equilibrium (Chapter 12) adds a rule for surprises:

- Pretend everyone makes tiny mistakes ("trembles")
- Then every info set is reached, so Bayes' rule always works
- Beliefs $=$ the limit as the mistakes shrink to zero

Facts: SE $\subseteq$ WSE $\cap$ SPNE. And an SE always exists

Not covered in this class. Kreps & Wilson (1982, *Econometrica*).

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Donut truck: SE doesn't help. Some tremble gives any μ

Not covered in this class. Kreps & Wilson (1982, *Econometrica*).

Back to Herding

Suppose everyone guesses their own ball's color... until it stops making sense

Player 2 sees "Blue", draws blue:

$$P(\text{Blue jar}) = \frac{(.6)(.6)}{(.6)(.6) + (.4)(.4)} = _____$$

Player 3 sees "Blue, Blue", draws **red**:

Two blue balls revealed, one red. $P(\text{Blue jar}) =$ _____ Guess? _____

Anderson & Holt (1997, *AER*). Not in the book.

Back to Herding

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Cascade: from now on, guesses reveal nothing. Everyone herds

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This is Bayesian updating **along the path of play**: guesses reveal balls

Our groups: _____ cascades, _____ on the wrong color

Anderson & Holt (1997, *AER*). Not in the book.

Beliefs After Surprises

Off-path beliefs matter whenever someone does something unexpected:

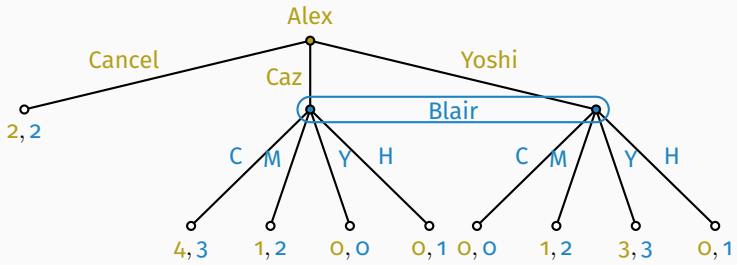
- **Entry:** a rival enters a market “nobody” would enter. Tough or desperate?
- **Recruiting:** a 5-star recruit decommits. Bad fit or a better offer?
- **Football:** going for it on 4th & 10 at your own 20. Genius or desperate?
- **Earnings:** a company delays its report. Bad news?
- **Friends:** your always-early friend is late. Traffic or trouble?

WSE: your answer has to be **a** belief, and your reaction has to fit it

Let’s Work One Out

Dead Phones, Again

Alex and Blair have dinner plans. Alex either **Cancels** (Blair knows), or goes to Cazuela's or Yoshi Sushi. Then both phones die



Blair can't tell where Alex went: Cazuela's (C), Massey's (M), Yoshi (Y), or Home (H). Massey's is halfway, so they'd find each other. Payoffs (Alex, Blair)

Dead Phones, Again

1. Write the strategic form. Find the pure NE. Any proper subgames?

2. Let $\mu =$ Pr(Cazuela's) at Blair's info set. Blair's expected payoff from C, M, Y, H?

3. Which NE are part of a WSE? With what beliefs?

4. If Alex mixes Cancel $\frac{1}{2}$, Cazuela's $\frac{1}{3}$, Yoshi $\frac{1}{6}$: Blair's belief? Best reply?

What Have We Learned, Charlie Brown?

- **Assessment** (σ, μ): strategies plus beliefs at every info set
- **Sequential rationality**: best choice at every info set, given beliefs
- **Bayesian updating**: beliefs from σ by Bayes' rule, when reachable
  - Unreached info sets: any beliefs
- **WSE** $=$ both. Kills Selten's horse's silly SPNE
- WSE $\Rightarrow$ NE, but not always SPNE. Fix: sequential equilibrium

Reading & Practice

- Bonanno, Chapter 11
  - §11.1–11.3: read. Skim the $\pi_i(H \mid \sigma, \mu)$ notation
- Optional: Chapter 12 for sequential equilibrium
- Exercises: §11.4.1 (11.1, 11.2), §11.4.2 (11.4), §11.4.3 (11.6, 11.7)
- Solutions in §11.5. Try them first